Probability Theory

Omkar Tasgaonkar
otasgaonkar@ucla.edu

These are my notes on probability theory, which I am studying for C271A and as an extension of what I learned in analysis. I hope to cover the relevant measure theory whenever possible, and I will also add completed practice problems as well.

Textbooks Used to Develop these Notes:

1. ***Probability: Theory and Examples*** - *Rick Durrett*
2. ***Real Analysis*** - *Gerald Folland*

Other Textbooks Referenced:

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NOTE: Any sections/subsections marked with an asterisk are extra sections and have been added to this document as extensions of introduced theory.

1 Measure Spaces

1.1 Algebras and Rings

We begin with some basic measure theory which will motivate the formulation of the σ-algebra and allow us to go from generality to specificity in defining probability spaces.

We begin by defining a probability space as a triple $(\Omega, \mathcal{F}, P)$. The first element Ω is called the set of outcomes or the "sample space", which is just an underlying set that can be defined as we wish.

Definition 1.1 (σ-algebra)**.** *The second element $\mathcal{F}$ is called a σ-algebra, satisfying the following properties:*

1. $\Omega \in \mathcal{F}$
2. $\forall S, P \in \mathcal{F} : S \setminus P \in \mathcal{F}$
3. *The countable (finite or countably infinite) union of sets in $\mathcal{F}$ is in $\mathcal{F}$:*

$$\forall \{S_n\}_{n\in\mathbb{N}} \in \mathcal{F}^{\mathbb{N}} : \bigcup_{n\in\mathbb{N}} S_n \in \mathcal{F}$$

We can equivalently replace this condition with closure under disjoint unions.

We call the set S an event (specifically an event with an assigned probability, which we shall see later). We also define an algebra over sets $\mathcal{A}$, which weakens condition 3 to finite unions.

Consider the following example of a set that is an algebra but not a σ-algebra, quite similar to the countable/co-countable σ-algebra. Define:

$$\mathcal{A} \coloneqq \{A \subseteq X : A \text{ finite} \vee A^c \text{ finite}\}$$

where X is infinite.

Lemma 1.1. *We can derive other properties of sets in the σ-algebra, such as:*

1. $\varnothing \in \mathcal{F}$
2. $\forall \{S_n\}_{n\in\mathbb{N}} \in \mathcal{F}^{\mathbb{N}}$*:*

$$\bigcap_{n\in\mathbb{N}} S_n \in \mathcal{F}$$

3. $\forall S \in \mathcal{F} : \Omega \setminus S \in \mathcal{F}$

These properties also hold for algebras in the finite sense.

Proof. **σ-algebras**: For $\varnothing \in \mathcal{F}$, this follows from the fact that $\Omega \in \mathcal{F}$ and from property two, $\Omega \setminus \Omega = \varnothing \in \mathcal{F}$.

The second statement follows from DeMorgan's Laws. If $S_n \in \mathcal{F}$, then $S_n^c \in \mathcal{F}$, which tells us that:

$$\bigcup_{n\in\mathbb{N}} S_n^c = \left(\bigcap_{n\in\mathbb{N}} S_n\right)^c \in \mathcal{F}$$

Once again by the completement axiom, we see that:

$$\bigcap_{n\in\mathbb{N}} S_n \in \mathcal{F}$$

As for the third statement, we know that since $\Omega \in \mathcal{F}$, then we have that for all $S \in \mathcal{F}$, that $\Omega \setminus S \in \mathcal{F}$. □

Proof. **Algebras**: $\varnothing \in \mathcal{F}$ follows from property 2.

Since $\Omega \in \mathcal{F}$, we have that $\forall S \in \mathcal{F} : \Omega \setminus S \in \mathcal{F}$. So, for any finite collection $\{S_i\}_{i=0}^n$, we have that $S_i^c \in \mathcal{F}$:

$$\bigcup_{i=0}^n S_i^c \in \mathcal{F} \implies \left(\bigcup_{i=0}^n S_i^c\right)^c = \bigcap_{i=0}^n S_i \in \mathcal{F}$$

□

Lemma 1.2. *The intersection of a family of σ-algebras for a set Ω is a σ-algebra.*

Proof. Let us call the family $\{\mathcal{F}_\alpha : \alpha \in I\}$ such that we have valid measurable spaces $(\Omega, \mathcal{F}_\alpha)$, then:

$$\mathcal{F} \coloneqq \bigcap_{\alpha\in I} \mathcal{F}_\alpha$$

is a valid sigma algebra such that $(\Omega, \mathcal{F})$ is a measurable space. We see that:

$$\forall \alpha \in I : \Omega \in \mathcal{F}_\alpha \implies \Omega \in \bigcap_{\alpha\in I} \mathcal{F}_\alpha$$

so property 1 in Definition 1.1 is fulfilled. Now, for some S in the intersection:

$$S \in \bigcap_{\alpha\in I} \mathcal{F}_\alpha \iff \forall \alpha \in I : S \in \mathcal{F}_\alpha$$

$$\implies \forall \alpha \in I : S^c \in \mathcal{F}_\alpha \iff S^c \in \bigcap_{\alpha\in I} \mathcal{F}_\alpha$$

thus property 2 is fulfilled. Now consider some $\{S_n\}_{n\in\mathbb{N}} \in \mathcal{F}^{\mathbb{N}}$, then:

$$\forall n \in \mathbb{N} : S_n \in \mathcal{F} \implies \forall \alpha \in I : S_n \in \mathcal{F}_\alpha$$

$$\implies \forall \alpha \in I : \{S_n\}_{n\in\mathbb{N}} \in \mathcal{F}_\alpha^{\mathbb{N}} \implies \forall \alpha \in I : \bigcup_{n\in\mathbb{N}} S_n \in \mathcal{F}_\alpha$$

$$\implies \bigcup_{n\in\mathbb{N}} S_n \in \mathcal{F}$$

so we have verified property 3. This is quite similar to topologies, where the intersection of topologies is a topology, but the union is not necessarily a topology. □

Consider some sample space $\Omega \coloneqq \{1, 2, 3, 4\}$, then consider:

$$\mathcal{F}_1 \coloneqq \{\Omega, \varnothing, \{1\}, \{2, 3, 4\}\}$$

$$\mathcal{F}_2 \coloneqq \{\Omega, \varnothing, \{2\}, \{1, 3, 4\}\}$$

both of which are σ-algebras. If we take the intersection:

$$\mathcal{F}_3 \coloneqq \mathcal{F}_1 \cap \mathcal{F}_2 = \{\Omega, \varnothing\}$$

which is the "coarsest" or trivial σ-algebra. Now taking the union:

$$\mathcal{F}_1 \cup \mathcal{F}_2 = \{\Omega, \varnothing, \{1\}, \{2\}, \{2,3,4\}, \{1,3,4\}\}$$

but by Definition 1.1 property 3, we would need $\{1,2\}$ to belong in the union, thus the union is not necessarily a σ-algebra.

An example of a σ-algebra is the countable σ-algebra, defined as:

$$\mathcal{F} \coloneqq \{S \subseteq \Omega : S \text{ countable} \vee S^c \text{ countable}\}$$

Proof. We have that $\Omega \in \mathcal{F}$ since $\varnothing$ is trivially countable. Now consider some $S \in \mathcal{F}$, then if S is countable, we know that $(S^c)^c$ is countable, thus $S^c \in \mathcal{F}$. If S is not countable, then we know that S^c must be countable, so $S^c \in \mathcal{F}$.

Now take a countable union of sets in the sequence $\{S_n\}_{n\in\mathbb{N}} \in \mathcal{F}^{\mathbb{N}}$. If each S_n is countable, then it follows that the union is in $\mathcal{F}$ from the fact that countable union of countable sets is countable. If each S_n is not countable, then:

$$\left(\bigcup_{n\in\mathbb{N}} S_n\right)^c = \bigcap_{n\in\mathbb{N}} S_n^c \in \mathcal{F}$$

since S_n not countable and $S_n \in \mathcal{F}$ means S_n^c is countable. Assume some are countable and some are not, then for the S_i that are not countable, we have that S_i^c is countable. This forces the complement of the union to be countable. □

Definition 1.2. *Given a space Ω and some set $S \subseteq \mathcal{P}(\Omega)$, we define the minimal algebra generated by S as:*

$$\mathcal{F}(S) \coloneqq \bigcap_{S\subseteq\mathcal{F}} \mathcal{F}$$

which is an algebra as per Lemma 1.2.

Consider two subsets $P \subseteq S \subseteq \mathcal{P}(\Omega)$, then we know that:

$$P \subseteq S \subseteq \mathcal{F}(S) \implies \mathcal{F}(P) \subseteq \mathcal{F}(S)$$

Since $\mathcal{F}(S)$ is a σ-algebra containing P thus belongs in the intersection of the minimal σ-algebra containing P. We say the Borel σ-algebra on $\mathbb{R}$ is one that is generated by the set of all open intervals:

$$I \coloneqq \{(a,b) : (\forall a, b \in \mathbb{R} : a < b)\} \subseteq \mathcal{P}(\mathbb{R})$$

$$\mathcal{B} \coloneqq \mathcal{F}(I)$$

The motivation of defining $\mathcal{B}$ as a minimal algebra of an algebra generated by open intervals is to essentially "hope" that unmeasurable/pathological sets like the Vitali set are eliminated from question. $\mathcal{B}$ will contain open intervals, closed intervals, and countable unions and intersections of them (so Cantor type sets are also included).

The issue is, while the Borel σ-algebra gives us candidates for what sets are measurable, it does not give the existence of a measure in $\mathbb{R}$.

Consider the minimal σ-algebra of an already existing σ-algebra $\mathcal{G}$, with the minimal σ-algebra being $\mathcal{F}(\mathcal{G})$. Then:

$$\mathcal{F}(\mathcal{G}) \coloneqq \bigcap_{\mathcal{G} \subseteq \mathcal{F}} \mathcal{F}$$

By definition, we have that $\mathcal{G} \subseteq \mathcal{F}(\mathcal{G})$. However, since $\mathcal{G} \subseteq \mathcal{G}$, then we have that:

$$\mathcal{F}(\mathcal{G}) \subseteq \mathcal{G} \implies \mathcal{F}(\mathcal{G}) = \mathcal{G}$$

We introduce a few more concepts:

Definition 1.3. *A ring $\mathcal{R} \subseteq \mathcal{P}(\Omega)$ is an algebera without the condition that $\Omega \in \mathcal{R}$. A σ-ring $\mathcal{R}_\sigma$ is a σ-algebra without the same condition.*

Lemma 1.3. *Properties of rings:*

1. *Rings (and respectively σ-rings) are closed under (countable) intersections*

Proof. **Property One:** Consider a ring $\mathcal{R}$ and a finite collection of sets $\{S_i\}_{i=0}^n \in \mathcal{R}^{n+1}$. We prove by induction, consider the base case $n = 1$. Then:

$$S_1 \cap S_0 = S_1 \setminus (S_1 \setminus S_0)$$

Since $S_1, S_0 \in \mathcal{R}$, then $S_1 \setminus S_0 \in \mathcal{R}$ and applying the set difference property again, we obtain that the intersection is in the ring. Now assume this to be true for some $n \in \mathbb{N}$. Then for $n+1$:

$$\bigcap_{i=0}^{n+1} S_i = S_{n+1} \cap \bigcap_{i=0}^{n} S_i = S_{n+1} \setminus \left(S_{n+1} \setminus \bigcap_{i=0}^{n} S_i \right) \in \mathcal{R}$$

Now consider a σ-ring $\mathcal{R}_\sigma$ and a sequence of sets $\{S_n\}_{n \in \mathbb{N}} \in \mathcal{R}_\sigma^{\mathbb{N}}$. We know that:

$$\bigcup_{n \in \mathbb{N}} S_n \in \mathcal{R}_\sigma$$

Now by the finite set difference property, we know that $E \setminus S_n \in \mathcal{R}_\sigma$, so their countable union is in the ring too. Now:

$$E \setminus \bigcup_{n \in \mathbb{N}} (E \setminus S_n) = E \cap \left(\bigcup_{n \in \mathbb{N}} (E \setminus S_n) \right)^c = E \cap \bigcap_{n \in \mathbb{N}} (E \setminus S_n)^c = E \cap \bigcap_{n \in \mathbb{N}} S_n$$

If we set $E = S_0$, then:

$$S_0 \cap \bigcap_{n \in \mathbb{N}} S_n = \bigcap_{n \in \mathbb{N}} S_n \in \mathbb{R}_\sigma$$

by once again applying the set difference property to E setminus the countable union (in the case that $E = S_0$). □

1.1.1 Problems (Folland and Durrett)

Folland 1.2.1D: If $\mathcal{R}_\sigma$ is a σ-ring, then $\mathcal{F} \coloneqq \{E \subseteq X : (\forall F \in \mathcal{R}_\sigma : E \cap F \in \mathcal{R}_\sigma)\}$ is a σ-algebra:

Proof. First we observe that X is in $\mathcal{F}$ since for all $F \in \mathcal{R}_\sigma$, we have that $X \cap F = F \in \mathcal{R}_\sigma$. Now, consider any $E, G \in \mathcal{F}$ such that:

$$\forall F \in \mathcal{R}_\sigma : E \cap F \in \mathcal{R}_\sigma \wedge G \cap F \in \mathcal{R}_\sigma$$

Then by set difference properties of the σ-ring, we have that:

$$(E \cap F) \setminus (G \cap F) \in \mathcal{R}_\sigma \implies (E \cap F) \cap (G^c \cup F^c) = E \cap G^c \cap F = (E \setminus G) \cap F \in \mathcal{R}_\sigma$$

thus $E \setminus G \in \mathcal{F}$. Finally, we must prove closure under countable union. Take a sequence $\{E_n\}_{n\in\mathbb{N}} \in \mathcal{F}^{\mathbb{N}}$ such that:

$$\forall F \in \mathcal{R}_\sigma, \forall n \in \mathbb{N} : E_n \cap F \in \mathcal{R}_\sigma$$

$$F \cap \bigcup_{n\in\mathbb{N}} E_n = \bigcup_{n\in\mathbb{N}} (F \cap E_n)$$

Since $F \cap E_n \in \mathcal{R}_\sigma$ for each E_n, then by closure under countable union of the σ-ring, we have that the countable union of E_n is in $\mathcal{F}$. Thus $\mathcal{F}$ is a σ-algebra. Here we used a σ-ring to describe a σ-algebra, a similar technique to what is used in Caratheodory's Extension Theorem. □

Folland 1.2.5: If $\mathcal{M}$ is the σ-algebra generated by $E \subseteq \mathcal{P}(X)$, then $\mathcal{M}$ is the union of the σ-algebras generated by $\mathcal{S}$ as $\mathcal{S}$ ranges over all countable subsets of E. In other words, we want to show:

$$\mathcal{F}(E) = \bigcup_{\alpha\in I} \mathcal{F}(S_\alpha)$$

Proof. Since $E \subseteq \mathcal{F}(E)$, we have that:

$$\forall \alpha \in I : S_\alpha \subseteq E \implies \forall \alpha \in I : \mathcal{F}(S_\alpha) \subseteq \mathcal{F}(E) \implies \bigcup_{\alpha\in I} \mathcal{F}(S_\alpha) \subseteq \mathcal{F}(E)$$

For the reverse inclusion, we show that the union is a sigma algebra itself. The space X is trivially in the union. If we take some P in the union, then:

$$\exists \alpha \in I : P \in \mathcal{F}(S_\alpha) \implies P^c \in \mathcal{F}(S_\alpha) \implies P^c \in \bigcup_{\alpha\in I} \mathcal{F}(S_\alpha)$$

Now consider some sequence $\{P_n\}_{n\in\mathbb{N}}$ made from elements in the union. Then we have that:

$$\forall n \in \mathbb{N}, \exists \alpha \in I : P_n \in \mathcal{F}(S_\alpha)$$

Denote each of these α as α_n based on the corresponding set P_n. Then we have that:

$$\bigcup_{n\in\mathbb{N}} S_{\alpha_n}$$

is a countable union of countable subsets of E and thus is a countable subset of E, so there must exist some α' such that:

$$\forall n \in \mathbb{N} : S_{\alpha_n} \subseteq \bigcup_{n\in\mathbb{N}} S_{\alpha_n} = S_{\alpha'} \implies \mathcal{F}(S_{\alpha_n}) \subseteq \mathcal{F}(S_{\alpha'})$$

Thus there exists an $\alpha' \in I$ such that:

$$\forall n \in \mathbb{N} : P_n \in \mathcal{F}(S_{\alpha'}) \subseteq \bigcup_{\alpha\in I} \mathcal{F}(S_\alpha)$$

so the union is a valid σ-algebra. Now if we define $E \coloneqq \{E_\beta : \beta \in J\}$, then we know that:

$$\forall \beta \in J, \exists \alpha'' \in I : E_\beta \in S_{\alpha''}$$

where the countable subset of E is simply $\{E_\beta\}$ for that specific $\beta \in J$. Thus:

$$E \subseteq \bigcup_{\alpha \in I} S_\alpha \subseteq \bigcup_{\alpha \in I} \mathcal{F}(S_\alpha) \implies \mathcal{F}(E) \subseteq \bigcup_{\alpha \in I} \mathcal{F}(S_\alpha)$$

where we once again applied the subset property and the fact that the σ-algebra generated by a σ-algebra is itself. Then if we take some $C \in \mathcal{F}(E)$, we have that:

$$\exists \alpha \in I : C \in \mathcal{F}(S_\alpha)$$

So any set in the σ-algebra generated by a possibly large (uncountable), complex set can be in the simpler σ-algebra generated by a countable subset of the set. □

Durrett 1.1.1: Let $\Omega = \mathbb{R}$, $\mathcal{F}$ be the σ-algebra of countable/co-countable sets, and $P(A)$ be 0 for a countable set and 1 for a cocountable set. Show that $(\Omega, \mathcal{F}, P)$ is a probability space.

Proof. We proved the countable/co-countable set is a σ-algebra, but we do it once again, this time using the set difference definition. We know that $\Omega \in \mathcal{F}$, since $\varnothing$ is countable.

Now consider any two subsets $E, F \in \mathcal{F}$. We can have the following situations:

1. E, F are both countable
2. E, F both have countable complements
3. E is countable and F has a countable complement
4. E has a countable complement and F is countable

In the first case, we have that $E \setminus F = E \cap F^c \subseteq E$, making it countable. The second case is similar, where $E \setminus F = E \cap F^c \subseteq F^c$, making it countable. This applies to the third case as well. For the fourth case $E \setminus F = E \cap F^c$. Its complement is $E^c \cup F$ which is countable, thus $E \setminus F \in \mathcal{F}$.

We use the same proof for countable unions, showing that $\mathcal{F}$ is a σ-algebra. Now since $\mathbb{R}$ is uncountable, we have that $P(\mathbb{R}) = 1$.

To prove countable additivity, assume each S_n in the sequence is countable and pairwise disjoint. Then the countable union is countable, so:

$$P\left(\bigcup_{n \in \mathbb{N}} S_n\right) = 0 = \sum_{n \in \mathbb{N}} P(S_n) = \sum_{n \in \mathbb{N}} 0$$

Now assume S_i, S_j have a countable complement such that $S_i \cap S_j = \varnothing$. Then:

$$S_i^c \cup S_j^c = \mathbb{R}$$

which is a contradiction since $\mathbb{R}$ is uncountable. So we can only have one such S_i in a disjoint collection with a countable complement (which by uncountability of reals is itself uncountable). In this case:

$$P\left(\bigcup_{n \in \mathbb{N}} S_n\right) = P\left(\bigcup_{n \neq i} S_n \cup S_i\right) = 1 = \sum_{n \neq i} 0 + 1$$

Since the union is uncountable, thus proving that $(\Omega, \mathcal{F}, P)$ is a probability space. □

Durrett 1.1.2: Show that $\mathcal{F}(S_d) = \mathcal{B}^d$ where $\mathcal{B}$ is the Borel set of $\mathbb{R}$. Where S_d is defined as:

$$S_d \coloneqq \{(a_1, b_1) \times ... \times (a_d, b_d) : (\forall i \in 1, ..., d : a_i < b_i)\}$$

Durrett 1.1.3:

Durrett 1.1.4: Show that if $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq ...$ are σ-algebras, then their countable union is an algebra. Given an example to show that the union need not be a σ-algebra.

Proof. We first see that the underlying space X is trivially in the union. Similarly, if:

$$P \in \bigcup_{n \in \mathbb{N}} \mathcal{F}_n \implies \exists n \in \mathbb{N} : P \in \mathcal{F}_n \implies P^c \in \mathcal{F}_n$$

Thus it is closed under complements. Now we must prove closure under finite unions. Consider a finite collection of elements in the union $\{P_i\}_{i=0}^n$, then:

$$\forall i \in 0, ..., n, \exists k \in \mathbb{N} : P_i \in \mathcal{F}_k$$

Since all the σ-algebras are nested, take $k(n) \coloneqq \max\{k \in \mathbb{N} : (\exists i \in 0, ..., n : P_i \in \mathcal{F}_k)\}$. In which case the finite union belongs to the union of the $\mathcal{F}_n$.

Consider the space $\mathbb{R}^{\mathbb{N}}$, the space of real-valued sequences. Then we have the nested σ-algebras:

$$\mathcal{F}(\{\{e_1\}\}) \subseteq \mathcal{F}(\{\{e_1\}, \{e_2\}\}) \subseteq ...$$

Observe that each of the σ-algebras under closure of unions only generate a finite set $\{e_1, ..., e_n\}$ (for the $n - th$ such σ-algebra). If we take the countable union of elements from the union:

$$E \coloneqq \bigcup_{n \in \mathbb{N}} \{e_n\} = \{e_1, e_2, ...\}$$

then we get the set of all canonical basis vectors, which cannot be contained in any of the σ-algebras. □

Durrett 1.1.5:

1.2 Measure

The third and final element of the triple is P, which we call the probability measure. A measure is defined as a following:

Definition 1.4. *$(X, \mathcal{F})$ is a measurable space on which we can put a measure $\mu : \mathcal{F} \to [0, \infty]$ with the properties:*

1. $\forall S \in \mathcal{F} : \mu(S) \geqslant 0 \wedge \mu(\varnothing) = 0$

   Note that we could also equivalently replace the second part with $\mu(\varnothing) \in [0, \infty)$.

2. *Given some $\{S_n\}_{n \in \mathbb{N}} \in \mathcal{F}^n$, if:*

$$\forall i \neq j : S_i \cap S_j = \varnothing \implies \mu\left(\bigcup_{n \in \mathbb{N}} S_n\right) = \sum_{n \in \mathbb{N}} \mu(S_n)$$

This second property will be quite important, and we shall call it countable additivity. Countable additivity gives a reason as to why we wanted property 3 in Definition 1.1, where:

$$\forall \{S_n\}_{n\in\mathbb{N}} \in \mathcal{F}^{\mathbb{N}} : \bigcup_{n\in\mathbb{N}} S_n \in \mathcal{F}$$

If this were not true, then we could have had some sequence of S_n in the σ-algebra where their countably infinite union is not measurable, so property 2 in this definition would not always hold.

Lemma 1.4. *This definition of a measure reveals additional properties:*

1. *Monotonicity:*
$$\forall A, B \in \mathcal{F} : A \subseteq B \implies \mu(A) \leqslant \mu(B)$$
2. *Subadditivity:*
$$\forall \{S_n\}_{n\in\mathbb{N}} \in \mathcal{F}^{\mathbb{N}} : \mu\left(\bigcup_{n\in\mathbb{N}} S_n\right) \leqslant \sum_{n\in\mathbb{N}} \mu(S_n)$$
3. *Continuity from below:*
$$\forall n \in \mathbb{N} : S_n \subseteq S_{n+1} \implies \mu\left(\bigcup_{n\in\mathbb{N}} S_n\right) = \lim_{n\to\infty} \mu(S_n)$$
4. *Continuity from above:*
$$\forall n \in \mathbb{N} : S_{n+1} \subseteq S_n \wedge \mu(S_0) < \infty \implies \mu\left(\bigcap_{n\in\mathbb{N}} S_n\right) = \lim_{n\to\infty} \mu(S_n)$$

Proof. Take the sets $A, B \setminus A$, then:

$$\mu(B) = \mu(A \cup (B \setminus A)) = \mu(A) + \mu(B \setminus A) \geqslant \mu(A)$$

thus proving the first property of non-negativity. Here we used countable additivity with the sequence $A, B \setminus A, \varnothing, \varnothing, ...$ to bring about finite additivity. This means the proof also requires that the measure $\mu(\varnothing) = 0$.

For the second property, we can rewrite the union as the following:

$$\bigcup_{n\in\mathbb{N}} S_n = \bigcup_{n\in\mathbb{N}} \left(S_n \setminus \bigcup_{i=0}^{n-1} S_i\right)$$

The reverse inclusion is trivial. For the forward inclusion, if x is in the union of S_n, then:

$$\exists n \in \mathbb{N} : x \in S_n$$

but it might be that x belongs to the intersection of multiple such sets, so by well ordering property and nonemptiness above, we can define:

$$m \coloneqq \min\{n \in \mathbb{N} : x \in S_n\}$$

So $x \in S_m$ which also tells us that for all $n \leqslant m$ that $x \notin S_n$, so:

$$\exists m \in \mathbb{N} : x \in S_m \setminus \bigcup_{i=0}^{m-1} S_i$$

We also have the added benefit that each set difference is disjoint from any other. So:

$$\mu\left(\bigcup_{n\in\mathbb{N}} S_n\right) = \mu\left(\bigcup_{n\in\mathbb{N}}\left(S_n \setminus \bigcup_{i=0}^{n-1} S_i\right)\right) = \sum_{n\in\mathbb{N}} \mu\left(S_n \setminus \bigcup_{i=0}^{n-1} S_i\right) \leqslant \sum_{n\in\mathbb{N}} \mu(S_n)$$

Now observe we can do the following rewrite:

$$\sum_{n\in\mathbb{N}} \mu\left(S_n \setminus \bigcup_{i=0}^{n-1} S_i\right) = \lim_{n\to\infty} \sum_{i=0}^{n} \mu\left(S_i \setminus \bigcup_{j=0}^{i-1} S_j\right)$$

Countable additivity gives us finite additivity (simply consider a sequence $A_0, A_1, A_2, ..., A_n, \varnothing, \varnothing, ...$), so we can rewrite the sum as:

$$= \lim_{n\to\infty} \mu\left(\bigcup_{i=0}^{n} S_i\right) = \mu\left(\bigcup_{n\in\mathbb{N}} S_i\right)$$

The reason why the third and fourth properties are dubbed "continuity" is because they essentially explain a limit exchange under certain sufficient conditions, where:

$$\mu\left(\bigcup_{n\in\mathbb{N}} S_i\right) = \mu\left(\lim_{n\to\infty} S_n\right) = \lim_{n\to\infty} \mu(S_n)$$

The proof follows simply from our observation before. Due to the nesting where $S_0 \subseteq S_1 \subseteq ...$, we have that:

$$\bigcup_{i=0}^{n} S_i = S_n$$

By the monotonicity property, we have that $\{\mu(S_n)\}_{n\in\mathbb{N}}$ is a non-decreasing sequence, so we can rewrite the limit as:

$$\lim_{n\to\infty} \mu(S_n) = \sup_{n\in\mathbb{N}} \mu(S_n)$$

For the fourth property, we use continuity from below. Consider a sequence $S_0 \supseteq S_1 \supseteq ....$ Then we have that $S_0 \setminus S_1 \subseteq S_0 \setminus S_2 \subseteq ...$, which tells us that:

$$\mu\left(\lim_{n\to\infty} S_0 \setminus S_n\right) = \lim_{n\to\infty} \mu(S_0 \setminus S_n)$$

Now observe:

$$\mu(S_0) = \mu(S_0 \setminus S_n \cup S_n) = \mu(S_0 \setminus S_n) + \mu(S_n) \implies \mu(S_0) - \lim_{n\to\infty} \mu(S_n) = \lim_{n\to\infty} \mu(S_0 \setminus S_n)$$

$$\implies \mu(S_0) - \lim_{n\to\infty} \mu(S_n) = \mu(S_0 \setminus \cap S_n) = \mu(S_0) - \mu\left(\bigcap_{n\in\mathbb{N}} S_n\right)$$

and via cancellation we have the desired property. Notice that the argument requires $\mu(S_0) < \infty$. If it were not less than infinity, then it may be possible that even the subsets of S_0 have infinite measure, which would mean:

$$\lim_{n\to\infty} \mu(S_n) = \infty$$

which would give an indeterminate subtraction on the left hand side. □

Furthermore, we can provide an equivalent definition for measure which does not require countable additivity:

Lemma 1.5. *The function μ is a measure if and only if it is non-negative (with measure of $\varnothing$ finite), finitely additive, and continuous from below.*

Proof. $\implies$: This direction was proved in Lemma 1.3.

$\impliedby$: Consider a sequence of pairwise disjoint sets $A_1, A_2, ...$ Then we have the sequence:

$$\left\{\bigcup_{i=0}^{n} A_n\right\}_{n\in\mathbb{N}}$$

which is non-decreasing (satisfying our conditiion for continuity from below). Then:

$$\mu\left(\bigcup_{n\in\mathbb{N}} A_n\right) = \mu\left(\bigcup_{n\in\mathbb{N}}\left(\bigcup_{i=0}^{n} A_n\right)\right) = \lim_{n\to\infty}\mu\left(\bigcup_{i=0}^{n} A_n\right)$$

By finite additivity, because the sequence is pairwise disjoint, we have that:

$$= \lim_{n\to\infty}\sum_{i=0}^{n}\mu(A_i) = \sum_{n\in\mathbb{N}}\mu(A_n)$$

Now we need to prove $\mu(\varnothing) = 0$. By finite additivity and $\mu(\varnothing) < \infty$:

$$\mu(\varnothing\cup\varnothing) = \mu(\varnothing) = \mu(\varnothing) + \mu(\varnothing) \implies \mu(\varnothing) = 0$$

So we must assume non-negativity (if we try to prove it via monotonicity, it becomes circular). Then we have that:

$$(\mu(\varnothing)\in[0,\infty)) + \text{finitely additive} + \text{continuous from below} \iff (\mu(\varnothing)=0) + \text{countably additive}$$

$$\iff (\mu(\varnothing)\in[0,\infty)) + \text{countably additive}$$

□

We can equivalently define a measure as:

Definition 1.5 (Alternative Definition of Measure)**.** *A measure $\mu : \mathcal{F} \to [0,\infty]$ has the properties:*

1. *For all finite collections of pairwise disjoint sets $S_0, S_1, ..., S_n$:*

$$\mu\left(\bigcup_{i=0}^{n} S_i\right) = \sum_{i=0}^{n}\mu(S_i)$$

2. *For all sequences $\{S_n\}_{n\in\mathbb{N}} \in \mathcal{F}^{\mathbb{N}}$:*

$$\forall n\in\mathbb{N} : S_n \subseteq S_{n+1} \implies \mu\left(\bigcup_{n\in\mathbb{N}} S_n\right) = \lim_{n\to\infty}\mu(S_n)$$

We can now make this notion more specific to probability and say:

Definition 1.6. *A probability measure is a function $P : \mathcal{F} \to [0, \infty]$ where:*

1. $P(\Omega) = 1$
2. *Given some sequence $\{S_n\}_{n\in\mathbb{N}} \in \mathcal{F}^{\mathbb{N}}$:*

$$\forall i \neq j : S_i \cap S_j = \varnothing \implies P\left(\bigcup_{n\in\mathbb{N}} S_n\right) = \sum_{n\in\mathbb{N}} P(S_n)$$

Lemma 1.6. *P is a measure on the measurable space $(\Omega, \mathcal{F})$ and:*

$$\forall S \in \mathcal{F} : P(\Omega \setminus S) = 1 - P(S)$$

Proof. Non-negativity and countable additivity follows by definition. We only need to prove that $\mu(\varnothing) = 0$. Simply take the sequence:

$$\Omega, \varnothing, \varnothing, ...$$

Then by countable additivity:

$$P(\Omega) = P(\Omega) + \sum_{n\in\mathbb{N}} P(\varnothing)$$

Since $P(\Omega) = 1$, we can apply cancellation, which forces $P(\varnothing) = 0$.

This is a sufficient condition for measure spaces in general. Given a non-negative function μ with countable additivity:

$$\exists S \in \mathcal{F} : (\mu(S) < \infty) + \text{countable additivity} \implies \mu(\varnothing) = 0$$

proving that μ is a measure on that measurable space.

We prove the second part now that we know that $P(\varnothing) = 0$. Because of this fact, we can say that countable additivity implies finite additivity, thus:

$$P(\Omega) = P(\Omega \setminus S \cup S) = P(\Omega \setminus S) + P(S)$$

$$\implies P(\Omega \setminus S) = 1 - P(S)$$

To do this proof, we require that $S \subseteq \Omega$ for the decomposition to hold. □

1.2.1 Problems

The Dirac Measure: Let X be the underling set and $\mathcal{M} = P(X)$ and some function $f : X \to [0, \infty]$, then we can define a measure:

$$\mu(E) \coloneqq \sum_{x\in E} f(x)$$

We then define the dirac meausre $\delta_{x_0}(E)$ by the fact that $f(x_0) = 1$ and $f(x) = 0$ for all $x \neq x_0$. In this case:

$$\forall E \in \mathcal{M} : \delta_{x_0}(E) \coloneqq \begin{cases} 0, & x_0 \notin E \\ 1, & x_0 \in E \end{cases}$$

We can easily check the null set has 0 measure and countable additivity holds (by the splitting of the sum for disjoint sets).

Folland 1.3.9: Given $(X, \mathcal{M}, \mu)$, then for $E, F \in \mathcal{M}$:

$$\mu(E) + \mu(F) = \mu(E \cup F) + \mu(E \cap F)$$

Proof.

$$\mu(F) = \mu(F \setminus E \cup (E \cap F)) = \mu(F \setminus E) + \mu(E \cap F)$$
$$\mu(E) = \mu(E \setminus F \cup (E \cap F)) = \mu(E \setminus F) + \mu(E \cap F)$$

Thus:

$$\mu(E) + \mu(F) = \mu(E \cap F) + \mu(F \setminus E) + \mu(E \setminus F) + \mu(E \cap F) = \mu(E \cap F) + \mu(E \cup F)$$

by finite additivity as $E \setminus F, F \setminus E, E \cap F$ are disjoint and their union is equal to the union of E and F.

This is familiar to the property of probability, where:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

□

1.3 Caratheodory Extension Theorem

We say that a null set $S \in \mathcal{F}$ is one which has measure 0. If we consider any subset $P \subseteq S$, then by monotonicity of the measure, we have that:

$$\mu(P) \leqslant \mu(S) = 0 \implies \mu(P) = 0$$

however, P may not be part of $\mathcal{F}$. We note an important property that the **COUNTABLE** union of null sets is a null set since:

$$\mu\left(\bigcup_{n\in\mathbb{N}} S_n\right) \leqslant \sum_{n\in\mathbb{N}} \mu(S_n) = 0$$

which follows from countably subadditivity of the measure (Lemma 1.4.2).

Definition 1.7. *A measure μ is complete on a measure space $(X, \mathcal{M}, \mu)$ if its associated σ-algebra contains all subsets of any null set and if the measure of those subsets is 0.*

We can extend the measure on the σ-algebra by the following:

Lemma 1.7. *Given a measure space $(X, \mathcal{F}, \mu)$, we say that $\overline{\mu}$ is a complete measure on $\overline{\mathcal{M}}$, where:*

$$\overline{\mathcal{M}} \coloneqq \{E \cup F : (E \in \mathcal{M} \wedge \exists N \in \mathcal{N} : F \subseteq N)\}$$

where $\mathcal{N}$ is the set of null sets in $\mathcal{M}$. We claim that $\overline{\mu}$ is a unique extension and that:

$$\overline{\mu}(E \cup F) = \mu(E)$$

$\overline{\mathcal{M}}$ is called the completion of $\mathcal{M}$.

Proof. We aim to show that $\overline{\mathcal{M}}$ is a σ-algebra, acting as the domain of $\overline{\mu}$. The space X is in the set.

Now consider some $E \cup F \in \overline{\mathcal{M}}$, we have that:

$$(E \cup F)^c = ((E \cup N) \cap (N^c \cup F))^c = (E \cup N)^c \cup (N^c \cup F)^c = (E \cup N)^c \cup (N \cap F^c)$$

$$= (E \cup N)^c \cup (N \setminus F)$$

Since $\mathcal{M}$ was a σ-algebra, we have that $(E \cup N)^c$ and $N \setminus F \subseteq N \in \mathcal{N}$, thus the complement belongs to $\overline{\mathcal{M}}$.

Now consider the countable union condition of some sequence $\{E_n \cup F_n\}$:

$$\bigcup_{n\in\mathbb{N}} (E_n \cup F_n) = \bigcup_{n\in\mathbb{N}} E_n \cup \bigcup_{n\in\mathbb{N}} F_n$$

Since $\mathcal{M}$ is a σ-algebra, we have that the union of all E_n belongs to it. Additionally by subadditivity of the measure:

$$\bigcup_{n\in\mathbb{N}} F_n \subseteq \bigcup_{n\in\mathbb{N}} N_n \wedge \mu\left(\bigcup_{n\in\mathbb{N}} N_n\right) \leqslant \sum_{n\in\mathbb{N}} \mu(N_n) = 0$$

Thus the union of null sets is a null set, so the union of F_n is a subset of a null set, which means the countable union of $E_n \cup F_n$ is the union of a set E in $\mathcal{M}$ and a set F in $\mathcal{N}$, proving closure under countable union.

We now need to show that $\overline{\mu} : \overline{\mathcal{M}} \to [0, \infty]$. First we show that it is a function such that:

$$E_1 \cup F_1 = E_2 \cup F_2 \implies \overline{\mu}(E_1 \cup F_1) = \overline{\mu}(E_2 \cup F_2) \iff \mu(E_1) = \mu(E_2)$$

Observe that:

$$E_1 \subseteq E_2 \cup F_2 \wedge E_2 \subseteq E_1 \cup F_1$$

which implies

$$\mu(E_1) \leqslant \mu(E_2) + \mu(N_2) = \mu(E_2)$$
$$\mu(E_2) \leqslant \mu(E_1) + \mu(N_1) = \mu(E_1)$$

We see that all subsets of null sets have measure zero as:

$$\forall F \subseteq N : \overline{\mu}(F) = \overline{\mu}(\varnothing \cup F) = \mu(\varnothing) = 0$$

It also trivially follows that the measure of $\varnothing$ under the complete measure is 0. Finally, for countable additivity, consider a sequence of $E_n \cup F_n$ pairwise disjoint. By the fact that $\overline{\mathcal{M}}$ is a σ-algebra:

$$\overline{\mu}\left(\bigcup_{n\in\mathbb{N}} (E_n \cup F_n)\right) = \overline{\mu}\left(\bigcup_{n\in\mathbb{N}} E_n \cup \bigcup_{n\in\mathbb{N}} F_n\right) = \mu\left(\bigcup_{n\in\mathbb{N}} E_n\right)$$

Since $\forall i \neq j : (E_i \cup F_i) \cap (E_j \cup F_j) = \varnothing$, we have that:

$$(E_i \cap E_j) \cup (E_i \cap F_j) \cup (E_j \cap F_i) \cup (F_i \cap F_j) = \varnothing$$

This tells us that all the intersections must be the empty set, so $E_i \cap E_j = \varnothing$, thus by countable additivity of μ:

$$\mu\left(\bigcup_{n\in\mathbb{N}} E_n\right) = \sum_{n\in\mathbb{N}} \mu(E_n) = \sum_{n\in\mathbb{N}} \overline{\mu}(E_n \cup F_n)$$

We finally need to show uniqueness. Assume there exists another such complete measure μ' such that:

$$\forall F \subseteq N : \mu'(F) = 0$$

so the two measures agree on subsets of null sets and on $\mathcal{M}$, in which case we want to show that:

$$\overline{\mu}(E \cup F) = \mu'(E \cup F)$$

We can write the following since $\mathcal{N} \subseteq \mathcal{M}$:

$$\mu'(E \cup F) \leqslant \mu'(E \cup N) = \overline{\mu}(E \cup N) \leqslant \overline{\mu}(E) \leqslant \overline{\mu}(E \cup F)$$

We can derive a similar inequality starting with $\overline{\mu}$, showing that the two agree on the completion of the σ-algebra. □

Definition 1.8. *Given a space X, we define an outer measure $\mu^* : \mathcal{P}(X) \to [0, \infty]$ with the following properties:*

1. $\mu^*(\varnothing) = 0$
2. $\forall A, B \in \mathcal{P}(X) : A \subseteq B \implies \mu^*(A) \leqslant \mu^*(B)$
3. $\forall \{A_n\}_{n\in\mathbb{N}} \in \mathcal{P}(X)^{\mathbb{N}} : \mu^* \left(\bigcup_{n\in\mathbb{N}} A_n\right) \leqslant \sum_{n\in\mathbb{N}} \mu^*(A_n)$

We require property two of monotonicity because we no longer have countable additivity to derive it. Instead we only have subadditivity.

We propose a specific formulation of the outer measure for some space X, where:

$$\forall A \in \mathcal{P}(X) : \mu^*(A) \coloneqq \inf \left\{ \sum_{n\in\mathbb{N}} \mu_0(E_n) : A \subseteq \bigcup_{n\in\mathbb{N}} E_n \right\}$$

Right now, μ_0 is a blackbox function. It maps some $E \subseteq \mathcal{P}(X)$ to $[0, \infty]$ such that $\mu_0(\varnothing) = 0$ and $X \in E$. This means for any set A in the powerset, the outer measure is always defined since a covering sequence where $E_n = X$ always exists.

Additionally by the positivity of μ_0 there is always a lower bound for the set, thus the infimum exists. We prove that this formulation is in actuality an outer measure:

Proof. We see that $\mu^*(\varnothing) = 0$, simply consider the covering sequence $\varnothing, \varnothing, ...$

For property 2, monotonicity, observe that for all covering sequences of the set B:

$$A \subseteq B \subseteq \bigcup_{n\in\mathbb{N}} E_n \implies \mu^*(A) \leqslant \sum_{n\in\mathbb{N}} \mu_0(E_n)$$

By properties of infimum, we have that $\mu^*(A) \leqslant \mu^*(B)$ (greatest lower bound).

Now consider the sequence of subsets in the powerset $\{A_n\}_{n\in\mathbb{N}}$. Then:

$$\forall n \in \mathbb{N} : A_n \subseteq \bigcup_{m\in\mathbb{N}} E_m^{(n)}$$

Now consider some value $\epsilon/2^n$, for which by definition of infimum, there exists some covering sequence for A_n such that:

$$\sum_{m\in\mathbb{N}} \mu_0(E_m^{(n)}) \leqslant \mu^*(A_n) + \frac{\epsilon}{2^n}$$

Additionally, we have that:

$$\bigcup_{n\in\mathbb{N}} A_n \subseteq \bigcup_{n\in\mathbb{N}} \bigcup_{m\in\mathbb{N}} E_m^{(n)}$$

where the covering sequence is countable (this is because there is a bijection between $\mathbb{N}^2$ and $\mathbb{N}$). So by infimum properties, we have that:

$$\begin{aligned} \mu^* \left(\bigcup_{n\in\mathbb{N}} A_n \right) &\leqslant \sum_{n\in\mathbb{N}} \sum_{m\in\mathbb{N}} \mu_0(E_m^{(n)}) \leqslant \sum_{n\in\mathbb{N}} \left(\mu^*(A_n) + \frac{\epsilon}{2^n} \right) \\ &\leqslant \sum_{n\in\mathbb{N}} \mu^*(A_n) + 2\epsilon \end{aligned}$$

Since this is true for all $\epsilon > 0$, we have proven countable subadditivity. Finite subadditivity follows from filling the remaining spots in the sequence of subsets with empty sets. □

Definition 1.9. *We say a set A is Caratheodory measurable or μ^*-measurable if:*

$$\forall E \in \mathcal{P}(X) : \mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

In proving a set A is μ^*-measurable, it suffices to prove:

$$\mu^*(E) \geqslant \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

As writing $E = E \cap (A \cup A^c) = (E \cap A) \cup (E \cap A^c)$, we obtain by subadditivity of the outer measure:

$$\mu^*(E) \leqslant \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

Theorem 1.1 (Caratheodory's Extension Theorem)**.** *If μ^* is an outer measure on some space X, then:*

$$\mathcal{M} \coloneqq \{A \subseteq X : (\forall E \subseteq X : \mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c))\}$$

is a complete σ-algebra, with $\mu^*|_{\mathcal{M}}$ being a complete measure.

Proof. We see that $X \in \mathcal{M}$ trivially. Additionally, we see that for any $A \in \mathcal{M}$, we have that $A^c \in \mathcal{M}$ by definition.

We now prove that $\mathcal{M}$ is closed under finite unions. Consider $A, B \in \mathcal{M}$. Then:

$$\mu^*(E \cap (A \cup B)) + \mu^*(E \cap A^c \cap B^c)$$

Simiar to Folland 1.3.9, we can write $A \cup B = (A \setminus B) \cup (A \cap B) \cup (B \setminus A)$, thus:

$$= \mu^*((E \cap A \cap B^c) \cup (E \cap A \cap B) \cup (E \cap B \cap A^c)) + \mu^*(E \cap A^c \cap B^c)$$

$$\leqslant \mu^*(E \cap A \cap B^c) + \mu^*(E \cap A \cap B) + \mu^*(E \cap B \cap A^c) + \mu^*(E \cap A^c \cap B^c)$$

Since A, B is μ^*-measurable:

$$= \mu^*(E \cap A) + \mu^*(E \cap A^c) = \mu^*(E)$$

Now that we have the property of finite unions, we can say $\mathcal{M}$ is an algebra. To upgrade it to a σ-algebra, we require closure under countable unions.

Now, we show that μ^* is finitely additive such that for some finite sequence of pairwise disjoint sets A_i in $\mathcal{M}$:

$$\mu^*\left(E \cap \bigcup_{i=0}^{n} A_i\right) = \sum_{i=0}^{n} \mu^*(E \cap A_i)$$

Consider the base case of $n = 1$, such that:

$$\mu^*(E \cap (A_0 \cup A_1)) = \mu^*((E \cap A_0 \cup E \cap A_1) \cap A_1) + \mu^*((E \cap A_0 \cup E \cap A_1) \cap A_1^c) = \mu^*(E \cap A_1) + \mu^*(E \cap A_0)$$

Now assume it holds true for some n, then for the $n+1$ case, since $A_{n+1} \in \mathcal{M}$:

$$\mu^*\left(\bigcup_{i=0}^{n+1} A_i\right) = \mu^*\left(\bigcup_{i=0}^{n} A_i \cup A_{n+1}\right)$$

$$= \mu^*\left(\left(\bigcup_{i=0}^{n} A_i \cup A_{n+1}\right) \cap A_{n+1}\right) + \mu^*\left(\left(\bigcup_{i=0}^{n} A_i \cup A_{n+1}\right) \cap A_{n+1}^c\right)$$

$$= \mu^*(A_{n+1}) + \mu^* \left(\bigcup_{i=0}^{n} A_i \right)$$

Nowe we can prove that $\mathcal{M}$ is closed under countable unions. Consider a sequence $\{A_n\}_{n\in\mathbb{N}} \in \mathcal{M}^{\mathbb{N}}$ and some $E \subseteq X$. By closure under finite unions:

$$\mu^*(E) = \mu^* \left(E \cap \bigcup_{i=0}^{n} \left(A_i \setminus \bigcup_{j=0}^{i-1} A_j \right) \right) + \mu^* \left(E \cap \bigcap_{i=0}^{n} A_i^c \right)$$

By finite additivity of the outer measure on the algebra $\mathcal{M}$, we have that:

$$\geqslant \sum_{i=0}^{n} \mu^* \left(E \cap A_i \setminus \bigcup_{j=0}^{i-1} A_j \right) + \mu^* \left(E \cap \bigcap_{n\in\mathbb{N}} A_n^c \right)$$

Since this is true for all $n \in \mathbb{N}$, we can take the limit as $n \to \infty$ and obtain:

$$\mu^*(E) \geqslant \sum_{n\in\mathbb{N}} \mu^* \left(E \cap A_n \setminus \bigcup_{i=0}^{n-1} A_i \right) + \mu^* \left(E \cap \bigcap_{n\in\mathbb{N}} A_n^c \right)$$

By subadditivity, we can recombine the sum and obtain:

$$\geqslant \mu^* \left(E \cap \bigcup_{n\in\mathbb{N}} A_n \right) + \mu^* \left(E \cap \bigcap_{n\in\mathbb{N}} A_n^c \right) \geqslant \mu^*(E)$$

We have shown an important equality, which we can state in full generality. Given some sequence of pairwise disjoint μ^*-measurable sets, we have that:

$$\mu^*(E) = \mu^* \left(E \cap \bigcup_{n\in\mathbb{N}} A_n \right) + \mu^* \left(E \cap \bigcap_{n\in\mathbb{N}} A_n^c \right) = \sum_{n\in\mathbb{N}} \mu^*(E \cap A_n) + \mu^* \left(E \cap \bigcap_{n\in\mathbb{N}} A_n^c \right)$$

Thus $\mathcal{M}$ is a σ-algebra. We now need to prove that μ^* is a measure on $\mathcal{M}$, for which it suffices to show countable additivity. Consider a disjoint sequence of sets $\{A_n\}_{n\in\mathbb{N}}$.

Then we have that their countable union is in the σ-algebra, so we can apply the measurability property:

$$\mu^* \left(\bigcup_{n\in\mathbb{N}} A_n \right) = \mu^* \left(\bigcup_{n\in\mathbb{N}} A_n \cap \bigcup_{n\in\mathbb{N}} A_n \right) + \mu^* \left(\bigcup_{n\in\mathbb{N}} A_n \cap \bigcap_{n\in\mathbb{N}} A_n^c \right)$$

$$= \sum_{n\in\mathbb{N}} \mu^*(A_n)$$

which follows from the highlighted equality.

Now we have that μ^* is a full measure on the σ-algebra $\mathcal{M}$. To show it is complete, consider a set E a subset of a null set N. Then taking any arbitrary $A \subseteq X$:

$$\mu^*(A) \leqslant \mu^*(A \cap E) + \mu^*(A \cap E^c) \leqslant \mu^*(N) + \mu^*(A \cap E^c) \leqslant \mu^*(A)$$

thus proving that $E \in \mathcal{M}$. By monotonicity of measure, we see that $\mu^*(E) = 0$. □

Definition 1.10 (Premeasure)**.** *We now introduce the concept of a premeasure $\mu_0 : \mathcal{A} \to [0, \infty]$ which is a non-negative function whose domain is an algebra such that:*

1. $\mu_0(\varnothing) = 0$
2. *For all sequences $\{A_n\}_{n\in\mathbb{N}} \in \mathcal{A}^{\mathbb{N}}$:*

$$\forall i \neq j : A_i \cap A_j = \varnothing \wedge \bigcup_{n\in\mathbb{N}} A_n \in \mathcal{A} \implies \mu_0\left(\bigcup_{n\in\mathbb{N}} A_n\right) = \sum_{n\in\mathbb{N}} \mu_0(A_n)$$

Observe that we place the additional sufficiency that the countable union must be in the algebra $\mathcal{A}$. This is not necessary for measures since their domains are σ-algebras which are automatically closed under countable unions, unlike algebras which are only closed under finite unions (and thus intersections).

Lemma 1.8. *A premeasure μ_0 is finitely subadditive and countable subadditive for collections whose unions belong in the algebra.*

Proof. Consider a finite collection of subsets in the algebra, $A_0, ..., A_n$. Then we can rewrite their union as a disjoint union:

$$\bigcup_{i=0}^{n} A_i = \bigcup_{i=0}^{n} \left(A_i \setminus \bigcup_{j=0}^{i-1} A_j \right)$$

Now by closure under countable union and finite set difference, we have each of the terms:

$$A_i \setminus \bigcup_{j=0}^{i-1} A_j \in \mathcal{A}$$

so the finite union of those terms is also in the algebra. By countable additivity (which reduces to finite additivity if we fill the remaining sequence with $\varnothing$), we have:

$$\mu_0\left(\bigcup_{i=0}^{n} A_i\right) = \sum_{i=0}^{n} \mu_0\left(A_i \setminus \bigcup_{j=0}^{i-1} A_j \right) \leqslant \sum_{i=0}^{n} \mu_0(A_i)$$

Monotonicity follows from non-negativity and finite additivity as we proved for measures. The proof for countable subadditivity is the same as long as the union belongs in the algebra. □

Lemma 1.9. *Given a space X and a premeasure $\mu_0 : \mathcal{A} \to [0, \infty]$, we have that:*

1. $\mu^*|_{\mathcal{A}} = \mu_0$
2. $\forall A \in \mathcal{A} : A$ *is μ^*-measurable*

where the outer measure μ^* is formulated as the infimum of sums of premeasures of covering sequences (view Definition 1.8).

Proof. **Part One:** If we consider some $A \in \mathcal{A}$, then by definition, we see that:

$$\mu^*(A) \leqslant \mu_0(A)$$

as $A, \varnothing, \varnothing, ...$ is a covering sequence for A. Now for the reverse inequality, since $\mu^*(A)$ exists, we have a covering sequence for A, such that:

$$A \subseteq \bigcup_{n\in\mathbb{N}} B_n \implies A = A \cap \bigcup_{n\in\mathbb{N}} B_n \in \mathcal{A}$$

We can rewrite A as a disjoint union:

$$A = \bigcup_{n\in\mathbb{N}} \left(A \cap B_n \setminus \bigcup_{i=0}^{n-1} B_i \right) \in \mathcal{A}$$

So by countable additivity and montonicity, we have that:

$$\mu_0(A) = \sum_{n\in\mathbb{N}} \mu_0 \left(A \cap B_n \setminus \bigcup_{i=0}^{n-1} B_i \right) \leqslant \sum_{n\in\mathbb{N}} \mu_0(A \cap B_n) \leqslant \sum_{n\in\mathbb{N}} \mu_0(B_n)$$

By the infimum definition of the outer measure μ^*, we see that $\mu_0(A) \leqslant \mu^*(A)$.

Part Two: We now want to show that:

$$\forall A \in \mathcal{A}, \forall E \subseteq X : \mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

We have that for all $\epsilon > 0$, there is some covering sequence of E such that:

$$\mu^*(E) + \epsilon \geqslant \sum_{n\in\mathbb{N}} \mu_0(B_n) = \sum_{n\in\mathbb{N}} \mu_0(B_n \cap (A \cup A^c))$$

$$= \sum_{n\in\mathbb{N}} \mu_0((B_n \cap A) \cup (B_n \cap A^c))$$

Now we see that $B_n \cap A \cap B_n \cap A^c = \varnothing$ for all $n \in \mathbb{N}$, so by finite additivity of the premeasure:

$$= \sum_{n\in\mathbb{N}} (\mu_0(B_n \cap A) + \mu_0(B_n \cap A^c))$$

Now since the sequence of B_n covered E, we have that the sequence of $B_n \cap A$ covers $E \cap A$. Similarly for $E \cap A^c$. Thus by definition of the outer measure:

$$\geqslant \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

Since this is true for all $\epsilon > 0$, we have that:

$$\mu^*(E) \geqslant \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

The reverse inequality follows from the finite subadditivity of the outer measure μ^*. □

Now since $\mathcal{A}$ is μ^*-measurable, we have that $\mathcal{A} \subseteq \mathcal{M}$ as defined in the Caratheodory Extension Theorem. Thus tells us that $\mathcal{F}(\mathcal{A}) \subseteq \mathcal{M}$.

As a result, we have that $\mu^*|_{\mathcal{F}(\mathcal{A})}$ is a measure (countable additivity follows from countable additivity on $\mathcal{M}$).

1.4 The Lebesgue Measure

Definition 1.11 (Semialgebra)**.** *A semialgebra $\mathcal{S} \subseteq \mathcal{P}(X)$ has the properties:*

1. $X \in \mathcal{S}$
2. *Closure under finite intersection:*

$$\forall \{A_i\}_{i=0}^n \in \mathcal{S}^{n+1} : \bigcap_{i=0}^n A_i \in \mathcal{S}$$

3. *Complements as disjoint finite unions:*

$$\forall A \in \mathcal{S}, \exists \{B_i\}_{i=0}^n \in \mathcal{S}^{n+1}, \forall i \neq j : B_i \cap B_j = \varnothing \wedge A^c = \bigcup_{i=0}^n B_i$$

Lemma 1.10. *We can generate an algebra from a semialgebra $\mathcal{A}(\mathcal{S})$, where:*

$$\mathcal{A}(\mathcal{S}) \coloneqq \left\{ \bigcup_{i=0}^n A_i : (\{A_i\}_{i=0}^n \in \mathcal{S}^{n+1} \wedge \forall i \neq j : A_i \cap A_j = \varnothing) \right\}$$

Proof. We prove $\mathcal{A}(S)$ is an algebra. Since $X \in \mathcal{S}$, we have that $X \in \mathcal{A}(\mathcal{S})$.

We aim to show that the algebra is closed under complements. Consider some $C \in \mathcal{A}(\mathcal{S})$, then:

$$C = \bigcup_{i=0}^n A_i$$

where each of the A_i subsets are pairwise disjoint. Now, taking the complement:

$$C^c = \bigcap_{i=0}^n A_i^c = \bigcap_{i=0}^n \left(\bigcup_{j=0}^m B_j^{(i)} \right)$$

where the union is a disjoint union (follows from Property 3 of semialgebras). When we expand the expression, we get unions of finite intersections of these $B_j^{(i)}$ sets (which belong in $\mathcal{S}$ by Property 2). By the distributive property of intersections over unions, if you take the intersection of any of two of these finite intersections, we see they are disjoint, thus $C^c \in \mathcal{A}(\mathcal{S})$.

Now, we can show closure under finite intersections, where:

$$\bigcap_{i=0}^n C_i = \bigcap_{i=0}^n \left(\bigcup_{j=0}^m A_j^{(i)} \right)$$

Once again distributing the intersection over the union, we get a union of finite intersections (which belong in $\mathcal{S}$), which we can prove is disjoint in the same manner. Combined with complement, it is closed under finite union.

□

2 Random Variables and Distributions